

Simulation Parameters Reference

Complete parameter list across every part of the system — engine, control logic, frontend, and analytics — with expected values and units. Defaults are reasonable starting points grounded in typical traffic engineering literature; tune once you have a working baseline.

1. Vehicle motion (IDM — Intelligent Driver Model)

Per-vehicle, applied every timestep.

Parameter	Symbol	Typical range	Default	Unit
Desired speed (free-flow)	v_0	8 - 20	13.9 ($\approx$ 50 km/h)	m/s
Desired time headway	T	1.0 - 2.5	1.5	s
Minimum gap at standstill	s_0	1.5 - 3.0	2.0	m
Max acceleration	$a_{\max}$	0.8 - 2.5	1.5	m/s ²
Comfortable braking deceleration	b	1.5 - 3.0	2.0	m/s ²
Acceleration exponent	δ	4 (fixed, rarely tuned)	4	—
Vehicle length	L	4.0 - 5.5	4.5	m
Reaction time (optional, adds realism)	τ	0 - 1.0	0.5	s

Notes: Higher T and s_0 = more cautious drivers, lower throughput. Lower $a_{\max}/b$ = sluggish, less realistic braking. If you want driver variability (some aggressive, some cautious), sample T , $a_{\max}$, b per vehicle from a normal distribution ($\pm$ 15-20% of default) rather than using one fixed value for the whole population.

2. Lane changing (simplified MOBIL)

Parameter	Typical range	Default	Unit
Politeness factor (weight on follower's discomfort)	0 - 1	0.3	—
Minimum safe deceleration for follower after lane change	2 - 4	3.0	m/s ²
Lane-change duration (visual only, not physics)	1 - 3	1.5	s
Minimum gap required in target lane	5 - 15	8	m

3. Vehicle spawning / arrivals

Parameter	Typical range	Default	Unit
Arrival rate (per approach)	100 - 1800	600	vehicles/hour
Arrival process	Poisson (random) or deterministic	Poisson	—
Number of approaches	2 - 4	4	—
Turning movement split (through / left / right)	sums to 100%	60 / 20 / 20	%
Peak/off-peak multiplier (optional)	0.5 - 2.0	1.0	× base rate

Notes: 600 veh/hr per approach is moderate urban traffic. Above ~1200 veh/hr per approach you'll likely see gridlock on a single-lane approach regardless of control type — useful for stress-testing but not a "normal" baseline.

4. Signal control

Parameter	Typical range	Default	Unit
Cycle length (total)	40 - 120	60	s
Green time per phase	15 - 45	25	s
Yellow time	3 - 5	3	s
All-red clearance interval	1 - 3	2	s
Number of phases	2 - 8	4 (NS-through, NS-left, EW-through, EW-left)	—
Actuated extension (if using actuated logic)	2 - 5	3	s per detected vehicle
Detector distance from stop line (actuated only)	20 - 50	30	m

Parameter	Typical range	Default	Unit
Min/max green (actuated only)	10 – 60	15 / 45	s

5. Roundabout control (gap acceptance)

Parameter	Typical range	Default	Unit
Critical gap (min acceptable gap to enter)	3.0 – 5.0	4.0	s
Follow-up time (gap between successive entering vehicles)	2.0 – 3.5	2.5	s
Circulating lane speed	15 – 30	25 (≈9 m/s)	km/h
Central island radius	5 – 15	8	m
Inscribed circle diameter (island + circulating lanes)	25 – 60	35	m
Number of circulating lanes	1 – 2	1	—
Entry lane count per approach	1 – 2	1	—
Deflection angle (entry path curvature, affects entry speed)	15° – 45°	30°	degrees

6. Hybrid model (stretch goal)

Parameter	Typical range	Default	Unit
Signal-controlled entry with roundabout circulation	boolean toggle	—	—
Metering signal cycle (entry only, not full intersection)	10 – 30	15	s
Applies only above a volume threshold	800 – 1500	1000	veh/hr total

7. Geometry / footprint (feeds the recommendation engine)

Parameter	Formula / range	Default	Unit
Signal footprint	$\text{intersection box} + \text{turn pockets} \approx \left(\frac{\text{lane_width}}{\text{lanes}} \right)^2 + \text{turn lane area}$	~600 - 900	m ²
Roundabout footprint	$\pi \times (\text{inscribed_radius})^2 + \text{approach tapers}$	~950 - 2800 (r=17.5m → ~962 m ²)	m ²
Lane width	3.0 - 3.7	3.5	m
Approach length modeled	100 - 300	150	m

8. Simulation engine / loop

Parameter	Typical range	Default	Unit
Timestep (Δt)	0.05 - 0.2	0.1	s
Run duration (single scenario view)	60 - 180	120	s
Warm-up period (excluded from metrics, lets queue stabilize)	10 - 30	20	s
Random seed	any int	fixed per comparison run	—

Notes: Keep the same random seed across signal/roundabout/hybrid runs being compared, so arrival patterns are identical and differences in results come from control logic, not random noise.

9. Frontend / rendering

Parameter	Typical range	Default	Unit
Scale factor	4 - 12	8	px per meter
Target frame rate	30 - 60	60	fps

Parameter	Typical range	Default	Unit
Playback speed multiplier	0.25 – 4	1.0	× real time
Vehicle render size	10×5 – 16×7	14×6	px

10. Metrics collected (output of every run)

Metric	Unit	Notes
Average wait time per vehicle	s	Time spent below ~1 m/s near the intersection
Average number of stops per vehicle	count	A "stop" = speed drops below ~0.5 m/s
Throughput	vehicles/min	Vehicles fully processed through the intersection
Max queue length	vehicles or m	Peak observed, per approach
Footprint	m ²	From Section 7
95th percentile wait time	s	Useful for "worst case" comparison, not just average

11. Recommendation engine weights (user-facing sliders)

Parameter	Range	Default	Unit
Weight: wait time	0 – 1	0.35	—
Weight: throughput	0 – 1	0.25	—
Weight: footprint/space	0 – 1	0.25	—
Weight: stops	0 – 1	0.15	—

Notes: Weights should be normalized to sum to 1 before computing the weighted score. Provide the default split above as a starting point; let the user drag sliders and re-normalize live.

12. Parameter sweep (batch analytics mode)

Sweep dimension	Suggested range for testing	Steps
Arrival rate	200 - 1400 veh/hr	5-7 steps
Lane count	1 - 2	2 steps
Roundabout radius	15 - 25 m	3 steps
Signal cycle length	40 - 100 s	3 steps

A full sweep across all four dimensions at these step counts is $5 \times 2 \times 3 \times 3 = 90$ runs per control type — trivial in batch mode (no rendering) even at $\text{dt} = 0.1$, likely seconds to low minutes total.